CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-501: Introduction to DBMS

Session: Comprehensive PYQ Bank Examination | **Paper Code:** 18021 | **Complete 100% Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION – A (VERY SHORT ANSWER QUESTIONS • 3 MARKS EACH • ALL 5 COMPULSORY)

Q1. Define Database Management System (DBMS) and its fundamental purpose. [3 Marks]

Answer:

A **Database Management System (DBMS)** is a specialized software package designed to define, construct, manipulate, and share structured databases among various users and applications.

> **Four Fundamental Purposes of DBMS:**
> 1. **Controlled Redundancy:** Integrates multiple logical files to eliminate duplicate data storage.
> 2. **Data Consistency & Integrity:** Enforces business constraints and ACID properties across operations.
> 3. **Concurrent Multi-User Access:** Allows hundreds of simultaneous users to access data without conflicts.
> 4. **Data Security & Recovery:** Provides role-based access control, transaction logs, and disaster recovery.

Q2. Distinguish between Simple and Composite Attributes with examples. [3 Marks]

Answer:

Attribute Type	Definition & Characteristic	Real-World Examples
Simple Attribute (Atomic)	Cannot be further divided into smaller independent sub-components. Holds a single atomic value.	`Roll_No`, `Age`, `Gender`, `Salary`
Composite Attribute	Can be divided into smaller sub-attributes, each having independent semantic meaning.	`Name` → (First, Middle, Last) `Address` → (Street, City, State, PIN)

Q3. Explain Referential Integrity Constraint. [3 Marks]

Answer:

Referential Integrity is a relational constraint between two relations (tables) maintained through a **Foreign Key**. It dictates that:

> A value in the Foreign Key column of the referencing table **MUST** either match an existing Primary Key value in the referenced table, or otherwise be `NULL`.

- **Prevents Orphan Records:** Prevents child records from pointing to non-existent parent rows.
- **Cascading Actions:**
  - `ON DELETE CASCADE`: Deleting a parent row automatically deletes all matching child rows.
  - `ON DELETE SET NULL`: Deleting a parent row resets foreign key fields in child rows to `NULL`.

Q4. Differentiate between DDL and DML commands with syntax. [3 Marks]

Answer:

Feature	DDL (Data Definition Language)	DML (Data Manipulation Language)
Primary Objective	Defines, alters, and drops database schema structures and tables.	Manages, inserts, updates, and retrieves data tuples within existing tables.
Transaction Behavior	**Auto-committed** immediately; changes cannot be rolled back via TCL.	**Not auto-committed** by default; can be committed or rolled back using `ROLLBACK`.
Key SQL Commands & Syntax	`CREATE TABLE Student(Id INT);` `ALTER TABLE Student ADD Age INT;` `DROP TABLE Student;` `TRUNCATE TABLE Student;`	`INSERT INTO Student VALUES(1, 20);` `UPDATE Student SET Age=21 WHERE Id=1;` `DELETE FROM Student WHERE Id=1;` `SELECT * FROM Student;`

Q5. Explain Functional Dependency (FD) and Multivalued Dependency (MVD). [3 Marks]

Answer:

1. Functional Dependency (X → Y):

A constraint between two sets of attributes X and Y in relation R. It states that for any two tuples t_1 and t_2, if $t_1[X] = t_2[X]$, then $t_1[Y] = t_2[Y]$ must also hold. Attribute set X uniquely determines attribute set Y.

Example: $Roll_No \rightarrow Student_Name, Branch$

2. Multivalued Dependency (X ↠ Y):

Occurs when presence of one or more rows in a table implies the presence of other rows. For a tuple (x, y, z), attribute set Y depends only on X and is completely independent of attribute set Z. MVD leads to **4NF decomposition**.

Example: In table (Course, Teacher, Book), if Course determines multiple Teachers and multiple Books independently: $Course \twoheadrightarrow Teacher$ *and* $Course \twoheadrightarrow Book$.

SECTION – B (SHORT ANSWER QUESTIONS • 7.5 MARKS EACH • ATTEMPT ANY TWO)

Q6. Discuss Anomalies associated with Relational Database (Insertion, Deletion, Update). [7.5 Marks]

Answer:

When relational schemas are unnormalized, redundant information causes three major operational anomalies:

Consider Unnormalized Relation: `Emp_Dept(EmpId, EmpName, DeptId, DeptName, DeptHOD)`

1. **Insertion Anomaly:**
   Occurs when certain facts cannot be recorded without prematurely adding unrelated attributes. For example, a new department (e.g., DeptId = 'D99', DeptName = 'AI Research') cannot be inserted until at least one employee is hired, because `EmpId` is the primary key and cannot accept `NULL`.
2. **Deletion Anomaly:**
   Occurs when deleting one piece of data inadvertently destroys critical unrelated information. For example, if employee 'E101' is the sole member of department 'D05', deleting employee 'E101' simultaneously deletes all records of department 'D05' (its name, budget, HOD) from the database.
3. **Update (Modification) Anomaly:**
   Occurs when data redundancy causes duplicate copies across multiple tuples. If department 'D01' has 500 employees, changing the department name requires updating all 500 rows. Missing even one row results in conflicting, inconsistent database states.

Solution: Normalization decomposes such relations into separate tables (e.g., `Employee` and `Department`) linked by foreign keys.

Q7. Explain Super Key, Candidate Key, and Primary Key with suitable examples. [7.5 Marks]

Answer:

Consider relation: `Student(RollNo, RegNo, Email, Name, Age, Phone)`

> **Hierarchical Relationship:** Super Key &supd; Candidate Key &supd; Primary Key

Key Category	Formal Definition & Properties	Examples from Student

Super Key	Any set of one or more attributes that uniquely identifies each tuple in a relation. May contain redundant attributes.	`{RollNo}`
		`{RegNo}`
		`{Email}`
		`{RollNo, Name}`
		`{RegNo, Age, Phone}`

Candidate Key	A **minimal** super key. Contains no extraneous attributes; removing any single attribute destroys unique identification.	`{RollNo}`
		`{RegNo}`
		`{Email}`

Primary Key	The single candidate key chosen by the database designer as the principal unique identifier. **Cannot be NULL** and must be unique.	`{RollNo}` (Chosen as primary key)

Q8. What is Normalization? Explain Second Normal Form (2NF) and Third Normal Form (3NF). [7.5 Marks]

Answer:

Normalization is a systematic technique for structuring relational tables to eliminate data redundancy, prevent insertion/deletion/update anomalies, and preserve data integrity.

> **1. Second Normal Form (2NF):**
>
> - **Condition 1:** Relation must already satisfy First Normal Form (1NF) — all attributes atomic.
> - **Condition 2: No Partial Dependency:** Every non-prime attribute must depend on the *entire* candidate key, not on any proper subset of a composite candidate key.
>
> *Example: In* `StudentCourse(StudentID, CourseID, Grade, CourseName)` *with candidate key* `{StudentID, CourseID}` *:*
>
> `CourseID → CourseName` is a partial dependency. Resolved by decomposing into: `Enrollment(StudentID, CourseID, Grade)` and `Course(CourseID, CourseName)`.

> **2. Third Normal Form (3NF):**
>
> - **Condition 1:** Relation must already be in 2NF.
> - **Condition 2: No Transitive Dependency:** Non-prime attributes must not depend on other non-prime attributes. For every non-trivial FD `X → Y`:
> - Either **X is a Super Key**, OR
> - **Y is a Prime Attribute** (part of any candidate key).
>
> *Example: In* `Employee(EmpId, EmpName, DeptId, DeptName)` *where* `EmpId → DeptId` *and* `DeptId → DeptName` *:*
> Decompose into `Emp(EmpId, EmpName, DeptId)` and `Dept(DeptId, DeptName)`.

SECTION – C (LONG ANSWER QUESTIONS • 15 MARKS EACH • ATTEMPT ANY THREE)

Q9. Define Boyce-Codd Normal Form (BCNF). Compare 3NF and BCNF with a suitable example. [15 Marks]

Answer:

Boyce-Codd Normal Form (BCNF) is a stricter, stronger version of 3NF. A relation R is in BCNF if and only if:

> For every non-trivial Functional Dependency `X → Y` in R, **X MUST be a Super Key**.

Key Difference between 3NF and BCNF:

Property	Third Normal Form (3NF)	Boyce-Codd Normal Form (BCNF)
Condition for X → Y	Valid if X is Super Key **OR** Y is a prime attribute.	Valid **ONLY** if X is a Super Key (no relaxation for prime Y).
Dependency Preservation	Always preserves all functional dependencies.	May occasionally sacrifice dependency preservation.
Redundancy Level	May retain slight redundancy for overlapping keys.	Strictly zero redundancy caused by functional dependencies.

Classic Example Illustrating 3NF vs BCNF:

Consider relation `StudentAdvisor(StudentID, Subject, Advisor)` with business rules:

- A student can take multiple subjects and have an advisor for each: `{StudentID, Subject} → Advisor`
- Each advisor teaches only one subject: `Advisor → Subject`
- Candidate Keys: `{StudentID, Subject}` and `{StudentID, Advisor}`.
- Prime Attributes: `StudentID`, `Subject`, `Advisor`.

> **Check for 3NF:** For FD `Advisor → Subject`, Advisor is not a super key, but **Subject is a prime attribute**. Therefore, the table is in **3NF** ✓.
>
> **Check for BCNF:** In `Advisor → Subject`, Advisor is **NOT** a super key. Hence, this table **violates BCNF** ✗.
>
> **BCNF Decomposition:** Split into two relations:
>
> 1. `AdvisorSubject(Advisor, Subject)` — Key: `Advisor` ✓
> 2. `StudentAdvisor(StudentID, Advisor)` — Key: `{StudentID, Advisor}` ✓

Q10. Explain Relational Algebra Fundamental Operations with examples. [15 Marks]

Answer:

Relational Algebra is a procedural query language consisting of operations that take one or two relations as input and produce a new relation as output.

Operation & Symbol	Description & Formal Rule	Concrete Example Expression
Selection (σ)	Selects tuples that satisfy a given predicate condition. Acts as a horizontal filter.	$\sigma_{Marks > 75}(Student)$ Filters students scoring above 75.
Projection (π)	Selects specified attribute columns and automatically discards duplicate tuples. Vertical filter.	$\pi_{RollNo, Name}(Student)$ Extracts RollNo and Name columns.
Cartesian Product (×)	Combines each tuple of relation R with every tuple of relation S. Output size: $\|R\| \times \|S\|$.	$Student \times Course$
Union (∪)	Combines unique tuples from two union-compatible relations (identical arity and compatible domains).	$\pi_{Name}(Faculty) \cup \pi_{Name}(Student)$
Set Difference (−)	Returns tuples present in relation R but not in relation S (must be union-compatible).	$\pi_{RollNo}(AllStudents) - \pi_{RollNo}(Placed)$
Rename (ρ)	Renames a relation or its attributes to prevent ambiguities in self-joins.	$\rho_{S1}(Student)$
Natural Join (⋈)	Equi-join combining Cartesian product, equality selection on common attributes, and duplicate column removal.	$Student \bowtie Department$

Q11. Explain SQL Characteristics and Five Aggregate Functions with SQL queries. [15 Marks]

Answer:

SQL Characteristics: SQL is a declarative, non-procedural language allowing users to specify *what* data is needed rather than *how* to retrieve it. It integrates DDL, DML, DCL, and TCL commands with full ACID compliance.

Five Core Aggregate Functions (operating on Employee table):

Function	Purpose	SQL Query Example & Description
`COUNT()`	Counts number of non-null values or total rows matching criteria.	`SELECT COUNT(*) FROM Employee WHERE Dept = 'IT';` Returns total number of employees in IT department.
`SUM()`	Calculates total arithmetic sum of numeric values.	`SELECT SUM(Salary) FROM Employee WHERE Dept = 'Sales';` Calculates total monthly payroll for Sales department.
`AVG()`	Calculates average mean value of numeric attribute.	`SELECT AVG(Salary) FROM Employee;` Returns average company-wide employee salary.
`MIN()`	Returns smallest minimum value in the attribute set.	`SELECT MIN(Salary) FROM Employee;` Finds the lowest compensation in the firm.
`MAX()`	Returns highest maximum value in the attribute set.	`SELECT Dept, MAX(Salary) FROM Employee GROUP BY Dept;` Returns highest salary earned in each department.

Q12. Explain Concurrency Control and Two-Phase Locking (2PL) Protocol with its pitfalls. [15 Marks]

Answer:

Concurrency Control coordinates simultaneous transaction execution to prevent conflicts (Lost Update, Uncommitted Read, Inconsistent Analysis) and ensure Serializable schedules.

Two-Phase Locking (2PL) Protocol:

1. **Growing (Expansion) Phase:** Transaction may acquire locks (Shared `S-Lock` or Exclusive `X-Lock`) but **cannot release any lock**.
2. **Lock Point:** The moment the transaction has acquired all necessary locks.
3. **Shrinking (Contraction) Phase:** Transaction may release locks but **cannot acquire any new locks**.

Theorem: Any schedule following the 2PL protocol is guaranteed to be **Conflict Serializable**.

Pitfalls of Basic 2PL & Solutions:

- **Cascading Aborts / Rollbacks:** If T1 releases a lock early in its shrinking phase and then aborts, any transaction T2 that read T1's dirty data must also abort.
  Solution → Strict 2PL: All Exclusive (X) locks must be held until the transaction commits or aborts.
- **Deadlocks:** 2PL does not prevent cyclic wait conditions where T1 holds Lock A waiting for Lock B, while T2 holds Lock B waiting for Lock A.
  Deadlock Handling Mechanisms:
  - *Deadlock Detection:* Maintains a dynamic **Wait-For Graph (WFG)**; cycles indicate deadlock and trigger victim abort.
  - *Deadlock Prevention:* Uses transaction timestamps: **Wait-Die** (non-preemptive) or **Wound-Wait** (preemptive).

Q13. Write detailed notes on: (a) Three-Schema Architecture, (b) Data Independence, and (c) Role of DBA. [15 Marks]

Answer:

(a) Three-Schema (ANSI/SPARC) Architecture:

1. **External Level (View Level):** Multiple tailored views for end-users and applications; conceals all irrelevant details and restricts unauthorized access.
2. **Conceptual Level (Logical Level):** Community-wide global schema describing what data is stored, entities, relationships, attributes, and constraints (tables, primary/foreign keys).
3. **Internal Level (Physical Level):** Physical data storage on disk, block allocation, record layout, B-Tree indexes, hashing, and encryption algorithms.

(b) Data Independence:

Independence Level	Definition & Mechanism	Real-World Scenario
Physical Data Independence	Ability to alter physical schema (disk storage, indexes, RAID, hashing) without modifying logical/conceptual schema.	Adding a B-Tree index on `Salary` to boost search speed without altering table structure.
Logical Data Independence	Ability to alter conceptual schema (add tables, add columns) without modifying existing external schemas or application programs.	Adding a `PassportNo` column to Student table without breaking existing grade-card report queries.

(c) Role and Responsibilities of Database Administrator (DBA):

- **Schema Definition & Storage Structure:** Creates logical tables and allocates physical disk storage paths.
- **Security & Authorization:** Grants and revokes permissions (`GRANT` , `REVOKE`) to prevent data leaks.
- **Performance Tuning:** Monitors slow queries, builds indexes, and reorganizes partitions.
- **Backup & Disaster Recovery:** Schedules incremental/full backups and handles crash recovery using transaction logs.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: Comprehensive PYQ Bank Examination | **Paper Code:** 18022 | **Complete 100% Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION – A (VERY SHORT ANSWER QUESTIONS • 3 MARKS EACH • ALL 5 COMPULSORY)

Q1. Why is Java called a Machine Independent language? [3 Marks]

Answer:

Java achieves architecture neutrality through its two-stage compilation and execution process:

> **Compilation & Execution Workflow:**
> `Source Code (.java) → javac Compiler → Intermediate Bytecode (.class) → JVM (Platform-Specific) → Machine Code`

- Java source code is compiled into intermediate binary **Bytecode** rather than platform-specific machine code.
- This bytecode can execute on any operating system (Windows, Linux, macOS) equipped with a suitable **Java Virtual Machine (JVM)**.
- This fulfills the foundational Java paradigm: **"Write Once, Run Anywhere" (WORA)**.

Q2. Explain the functionality and architecture of JVM. [3 Marks]

Answer:

The **Java Virtual Machine (JVM)** is an abstract computing machine that provides the runtime execution environment for compiled Java bytecode. It consists of three primary subsystems:

Subsystem	Components & Core Responsibilities
1. Class Loader Subsystem	Responsible for **Loading** (.class files), **Linking** (Verification, Preparation of static memory, Resolution), and **Initialization** of static blocks and variables.
2. Runtime Data Areas	• **Method Area:** Stores class metadata, bytecode, static variables. • **Heap Area:** Stores all dynamically instantiated objects and arrays. • **JVM Stack:** Stores stack frames for method calls and local variables. • **PC Registers:** Holds memory address of currently executing JVM instruction. • **Native Method Stack:** Holds native C/C++ function call frames.
3. Execution Engine	• **Interpreter:** Reads and executes bytecode instructions line by line. • **JIT (Just-In-Time) Compiler:** Compiles frequently executed "hotspots" directly into native machine code to dramatically improve speed. • **Garbage Collector (GC):** Automatically deallocates unreferenced heap memory.

Q3. Explain the 'super' keyword in Java with examples. [3 Marks]

Answer:

In Java, `super` is a reference variable used by a subclass to refer directly to its immediate parent (superclass) object.

Three Distinct Usages of 'super':

1. **Refer to Hidden Parent Field:** `super.variableName;` resolves variable name collisions when the subclass shadows a parent field.
2. **Invoke Overridden Parent Method:** `super.methodName();` allows a subclass method to explicitly invoke the original parent implementation.
3. **Invoke Parent Constructor:** `super();` or `super(args);` calls the parent class constructor (must strictly be the **first statement** in the subclass constructor).

Q4. Differentiate between Process-based and Thread-based Multitasking. [3 Marks]

Answer:

Comparison Criteria	Process-based Multitasking	Thread-based Multitasking
Definition & Scope	Executing two or more independent heavyweight programs concurrently (e.g., MS Word and Browser).	Executing two or more concurrent paths of execution within the *same* program.
Memory Allocation	Each process is allocated its own distinct, isolated private memory address space by the OS.	All threads belonging to the same process share identical memory address space and Heap memory.
Context-Switch Overhead	High overhead; OS must save registers, memory maps, and page tables.	Very low overhead; fast CPU register swap within shared process memory.
Communication Cost	Inter-Process Communication (IPC) is heavy and requires sockets, pipes, or shared files.	Inter-Thread Communication is fast and efficient using `wait()`, `notify()`, and shared variables.

Q5. Explain Anchor tag (<a>) and Table tag (<table>) in HTML. [3 Marks]

Answer:

- **Anchor Tag (`<a>`):** Creates clickable hyperlinks to navigate to external web pages, files, or internal bookmarks.

```
<a href="https://example.com/syllabus.html" target="_blank" title="View Syllabus">Download Syllabus</a>
```

  Attributes: `href` (target destination URL), `target="_blank"` (opens in new tab), `download` (prompts download).

- **Table Tag (`<table>`):** Organizes complex information into rows and columns.

```
<table border="1" cellpadding="6" cellspacing="0"> <tr><th>Roll No</th><th>Student Name</th> <th>Marks</th></tr> <tr><td>101</td><td>Amit Sharma</td><td>88</td></tr> </table>
```

  Tags: `<tr>` (table row), `<th>` (header cell, bold and centered), `<td>` (data cell), `colspan` / `rowspan` (cell merging).

SECTION – B (SHORT ANSWER QUESTIONS • 7.5 MARKS EACH • ATTEMPT ANY TWO)

Q6. What is an Exception? How do you handle multiple exceptions and use the finally block? [7.5 Marks]

Answer:

An **Exception** is an abnormal event occurring during program execution that disrupts the normal flow of instructions. Java provides robust exception handling using `try`, `catch`, `finally`, `throw`, and `throws`.

Rules for Handling Multiple Exceptions: Specific subclasses of `Exception` must appear *before* generic parent exceptions; otherwise a compilation error ("unreachable code") occurs.

Complete Working Java Code:

```
public class MultiCatchFinallyDemo { public static void main(String[] args) { try { int[] arr = {10, 20, 30}; int divisor = Integer.parseInt(args[0]); // May throw ArrayIndexOutOfBounds or NumberFormat int result = arr[1] / divisor; // May throw ArithmeticException if divisor == 0 System.out.println("Computed Result: " + result); } catch (ArithmeticException e) { System.err.println("Error: Cannot divide by zero! " + e.getMessage()); } catch (ArrayIndexOutOfBoundsException e) { System.err.println("Error: Please provide a command-line argument!"); } catch (NumberFormatException e) { System.err.println("Error: Input must be a valid integer!"); } catch (Exception e) { System.err.println("Generic Fallback Exception caught: " + e); } finally { // The finally block is GUARANTEED to execute regardless of whether an exception occurred or was handled System.out.println("Finally block executed: Cleaned up system resources and closed streams."); } } }
```

Q7. Explain Applet Life Cycle and write a program to display an image. [7.5 Marks]

Answer:

An **Applet** is a special Java program embedded in a web page and executed inside an applet viewer or Java-enabled browser. Its life cycle is managed by four lifecycle methods in `java.applet.Applet` plus the `paint()` method in `java.awt.Component`:

Applet Life Cycle Stages:

`init()` → `start()` → `paint(Graphics g)` ↔ `stop()` → `destroy()`

- `init()`: Called once upon loading; initializes variables, fonts, and images.
- `start()`: Called after `init()` and whenever the browser revisits the page; starts animation/threads.
- `paint(Graphics g)`: Redraws applet UI whenever screen refresh is required.
- `stop()`: Called when user navigates away; suspends background threads.
- `destroy()`: Called prior to unloading; cleans up memory resources.

Applet Code to Load and Display an Image:

```
import java.applet.Applet; import java.awt.Graphics; import java.awt.Image; /* <applet code="ImageDisplayApplet.class" width="400" height="300"></applet> */ public class ImageDisplayApplet extends Applet { private Image img; @Override public void init() { // Load image from the directory hosting the HTML document img = getImage(getDocumentBase(), "logo.png"); } @Override public void paint(Graphics g) { // Draw image at (x=30, y=30) with original dimensions g.drawImage(img, 30, 30, this); g.drawString("CCSU BCA 5th Sem: Java Applet Image Rendering", 30, 260); } }
```

Q8. Explain Threads and two ways to create Threads in Java. [7.5 Marks]

Answer:

A **Thread** is a lightweight sub-process representing the smallest unit of dispatchable execution. In Java, multithreading is achieved through two standard mechanisms:

Approach 1: Extending the `java.lang.Thread` Class

```
class WorkerThread extends Thread { @Override public void run() { for (int i = 1; i <= 3; i++) { System.out.println("WorkerThread executing step " + i + " on: " + Thread.currentThread().getName()); } } } public class ThreadDemo1 { public static void main(String[] args) { WorkerThread t1 = new WorkerThread(); t1.setName("Thread-Alpha"); t1.start(); // Spawns new OS thread and invokes run() } }
```

Approach 2: Implementing the `java.lang.Runnable` Interface (Recommended)

```
class WorkerTask implements Runnable { @Override public void run() { for (int i = 1; i ⩽ 3; i++) { System.out.println("RunnableTask step " + i + " on: " + Thread.currentThread().getName()); } } } public class ThreadDemo2 { public static void main(String[] args) { WorkerTask task = new WorkerTask(); Thread t2 = new Thread(task, "Thread-Beta"); t2.start(); } }
```

Key Advantage of Runnable: Since Java allows single class inheritance, implementing `Runnable` frees the class to inherit from another domain superclass while still enabling multithreading.

SECTION – C (LONG ANSWER QUESTIONS • 15 MARKS EACH • ATTEMPT ANY THREE)

Q9. Illustrate Inheritance. Why is multiple inheritance not possible in Java? How is it implemented via Interfaces? [15 Marks]

Answer:

Inheritance is an Object-Oriented mechanism whereby a subclass automatically acquires fields and methods from a superclass using the `extends` keyword, fostering code reusability and polymorphism.

The Diamond Problem (Why Multiple Class Inheritance is Prohibited):
If Class C inherits from both Class A and Class B, and both A and B define method `void display()` with different implementations: which version does an instance of Class C inherit? This ambiguity is known as the **Deadly Diamond of Death**. To avoid architectural complexity and pointer issues, Java designers disallowed multiple inheritance of classes.

Resolution using Interfaces:
Since interfaces contain only method declarations without state variables, there is zero ambiguity. A single Java class can implement multiple interfaces simultaneously.

```
interface Printable { void print(); // Abstract contract } interface Showable { void show(); } // Class implementing multiple interfaces class DocumentReport implements Printable, Showable { @Override public void print() { System.out.println("Printing document paper copy."); } @Override public void show() { System.out.println("Showing document on digital screen."); } } public class MultipleInheritanceDemo { public static void main(String[] args) { DocumentReport doc = new DocumentReport(); doc.print(); doc.show(); } }
```

Q10. What is JDBC? Explain types of JDBC drivers. Write a program to insert and select records using PreparedStatement. [15 Marks]

Answer:

Java Database Connectivity (JDBC) is a standard Java API (`java.sql` package) providing uniform, vendor-neutral database communication.

Driver Category	Architecture & Mechanism	Pros & Cons
Type-1: JDBC-ODBC Bridge	Translates JDBC calls into ODBC calls; requires native ODBC client installed.	Slow, platform-dependent, deprecated in modern Java.
Type-2: Native-API Driver	Converts JDBC calls directly into native C/C++ vendor client library calls.	Faster than Type-1, but requires native vendor client binaries on each machine.
Type-3: Network-Protocol Driver	Sends JDBC calls to an intermediate application server middleware.	Flexible, multi-database support, but adds server hop.
Type-4: Thin Driver (Pure Java)	Directly converts JDBC calls into database native network socket protocol.	**Highest performance**, 100% pure Java, platform independent, no client installation.

Complete JDBC Program (PreparedStatement Insert & Select):

```
import java.sql.Connection; import java.sql.DriverManager; import java.sql.PreparedStatement;
import java.sql.ResultSet; public class JdbcPreparedStatementDemo { public static void
main(String[] args) { String url = "jdbc:mysql://localhost:3306/college_db"; String user = "root";
String pass = "password"; try (Connection con = DriverManager.getConnection(url, user, pass)) { //
1. Parameterized INSERT using PreparedStatement (Prevents SQL Injection) String insertSql = "INSERT
INTO students (roll_no, name, marks) VALUES (?, ?, ?)"; try (PreparedStatement psInsert =
con.prepareStatement(insertSql)) { psInsert.setInt(1, 105); psInsert.setString(2, "Vikas Verma");
psInsert.setDouble(3, 91.5); int rows = psInsert.executeUpdate(); System.out.println("Rows
successfully inserted: " + rows); } // 2. Querying data using PreparedStatement String selectSql =
"SELECT roll_no, name, marks FROM students WHERE marks ⩾ ?"; try (PreparedStatement psSelect =
con.prepareStatement(selectSql)) { psSelect.setDouble(1, 75.0); ResultSet rs =
psSelect.executeQuery(); System.out.println("--- High Performing Students ---"); while (rs.next())
{ int rno = rs.getInt("roll_no"); String sname = rs.getString("name"); double scr =
rs.getDouble("marks"); System.out.println("Roll: " + rno + " | Name: " + sname + " | Marks: " +
scr); } } } catch (Exception e) { e.printStackTrace(); } } }
```

Q11. Explain Servlets and their Life Cycle. Compare Servlets and JSP. [15 Marks]

Answer:

A **Servlet** is a server-side Java class managed by a Servlet Container (e.g., Apache Tomcat) that receives client HTTP requests, processes business logic, and generates dynamic web responses.

Servlet Life Cycle (javax.servlet.Servlet Interface):

1. **Loading & Instantiation:** Servlet class bytecode is loaded and a single instance is instantiated in container memory.
2. **Initialization (`init(ServletConfig config)`):** Invoked **only once** immediately after instantiation; loads database connections and parameters.
3. **Request Servicing (`service(ServletRequest, ServletResponse)`):** Dispatches each incoming request thread to handler methods (`doGet()` or `doPost()`). Runs concurrently on separate threads.
4. **Destruction (`destroy()`):** Invoked **only once** when container shuts down or reloads; deallocates resources.

Comparison: Servlets vs JavaServer Pages (JSP):

Feature	Servlets	JSP (JavaServer Pages)
Core Paradigm	Pure Java code with embedded HTML written inside `out.println()` statements.	Standard HTML markup with embedded Java scriptlet tags (`<% ... %>`).
MVC Architectural Role	Best suited as **Controller** to manage request dispatching, validation, and business logic.	Best suited as **View** layer to present and format UI output for end users.
Compilation Speed	Pre-compiled into bytecode; executes rapidly from the very first request.	First-request is slower because JSP is internally translated into a Servlet and compiled.

Q12. Explain TCP/IP Socket Programming in Java with a client-server example. [15 Marks]

Answer:

Java Socket programming provides process-to-process communication across networks using the `java.net` package. TCP provides reliable, stream-oriented, full-duplex transmission.

Core Classes:

- `java.net.ServerSocket` : Listens on a dedicated port and waits for incoming client requests via `accept()` .
- `java.net.Socket` : Connects to a remote server IP and port; provides `getInputStream()` and `getOutputStream()` .

Server Application (ServerApp.java):

```
import java.io.*; import java.net.*; public class ServerApp { public static void main(String[]
args) throws IOException { int port = 5000; try (ServerSocket serverSocket = new
ServerSocket(port)) { System.out.println("Server listening on port " + port + "... Waiting for
client..."); try (Socket socket = serverSocket.accept()) { // Blocks until client connects
System.out.println("Client connected: " + socket.getInetAddress()); BufferedReader in = new
BufferedReader(new InputStreamReader(socket.getInputStream())); PrintWriter out = new
PrintWriter(socket.getOutputStream(), true); String clientMessage = in.readLine();
System.out.println("Received from client: " + clientMessage); out.println("Echo from Server: Hello,
" + clientMessage.toUpperCase()); } } } }
```

Client Application (ClientApp.java):

```
import java.io.*; import java.net.*; public class ClientApp { public static void main(String[]
args) throws IOException { String host = "127.0.0.1"; int port = 5000; try (Socket socket = new
Socket(host, port)) { PrintWriter out = new PrintWriter(socket.getOutputStream(), true);
BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()));
out.println("CCSU BCA 5th Sem Student"); String response = in.readLine();
System.out.println("Server response: " + response); } } }
```

Q13. Discuss Layout Managers in Java AWT/Swing (FlowLayout, BorderLayout, GridLayout, CardLayout, GridBagLayout). [15 Marks]

Answer:

A **Layout Manager** automatically computes the geometric sizing and spatial positioning of GUI components inside a container (such as `JFrame` or `JPanel`), abstracting away OS platform differences.

Layout Manager	Spatial Arrangement Strategy	Default Container & Behavior
FlowLayout	Arranges components sequentially in a horizontal line. When row width is exhausted, wraps automatically to next line.	Default for `JPanel`. Retains preferred component size.
BorderLayout	Divides container into 5 designated regions: `NORTH`, `SOUTH`, `EAST`, `WEST`, and `CENTER`. Components stretch to fill their region.	Default for `JFrame` and `JDialog`. Unused regions collapse into `CENTER`.
GridLayout	Arranges components in a strict two-dimensional matrix of equal-sized rectangular grid cells (rows × columns).	All components forced to identical width and height regardless of preferred size.
CardLayout	Stacks components on top of each other like a deck of playing cards. Only one component (card) is visible at a time.	Ideal for wizard installers, multi-tab dialogues, and step-by-step forms.
GridBagLayout	The most flexible and sophisticated layout manager. Components can vary in size and span multiple rows/columns using `GridBagConstraints`.	Allows precise, enterprise-grade responsive form alignment without nested panels.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: Comprehensive PYQ Bank Examination | **Paper Code:** 18023 | **Complete 100% Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION – A (VERY SHORT ANSWER QUESTIONS • 3 MARKS EACH • ALL 5 COMPULSORY)

Q1. Describe the various components of Data Communication. [3 Marks]

Answer:

Data communication is the exchange of data between two devices via a transmission medium. It comprises five essential components:

Component	Description & Role
1. Message	The information or payload to be communicated (Text, Numbers, Pictures, Audio, Video).
2. Sender (Transmitter)	The source device originating and transmitting the data message (PC, Workstation, Smartphone).
3. Receiver	The destination device that receives the transmitted data (Server, Printer, Mobile terminal).
4. Transmission Medium	The physical communication conduit carrying the signal from sender to receiver: Guided (Twisted pair, Coaxial cable, Fiber optic) or Unguided (Radio waves, Microwaves, Infrared).
5. Protocol	A formal set of governing rules that coordinate data transmission syntax, semantics, and timing (e.g., TCP/IP, HTTP).

Q2. Define Bandwidth and Network Congestion. [3 Marks]

Answer:

1. Bandwidth:
The maximum data transfer capacity of a communication channel over a given interval. In digital systems, it is measured in **bits per second (bps, Mbps, Gbps)**; in analog systems, it represents the frequency range (in Hertz, Hz) between upper and lower cutoff frequencies ($\texttt{Bandwidth} = f_{high} - f_{low}$).

2. Network Congestion:
A state where network nodes (routers/switches) carry more data packets than they can accommodate. When offered traffic load exceeds channel capacity, router queues overflow, causing **packet queuing delay, packet loss, and throughput collapse**.

Q3. Distinguish between MAC Address and IP Address. [3 Marks]

Answer:

Parameter	MAC Address (Physical Address)	IP Address (Logical Address)
OSI Layer	Data Link Layer (Layer 2).	Network Layer (Layer 3).
Address Length & Format	**48 bits (6 bytes)**; written in 12 hexadecimal digits (e.g., `00:1A:2B:3C:4D:5E`).	**32 bits** in IPv4 (e.g., `192.168.1.1`); **128 bits** in IPv6 (hexadecimal).
Permanence & Assignment	Permanently hard-coded into Network Interface Card (NIC) by manufacturer (OUI).	Dynamically (DHCP) or statically assigned based on network location.
Scope of Utility	Local node-to-node frame delivery within the same LAN segment.	Global end-to-end packet routing across multiple networks/Internet.

Q4. Differentiate between a Hub and a Switch. [3 Marks]

Answer:

Feature	Network Hub	Network Switch
Operating OSI Layer	Physical Layer (Layer 1).	Data Link Layer (Layer 2) / Multilayer (Layer 3).
Data Transmission Mode	**Broadcasts** all incoming frames to every connected port indiscriminately.	**Unicasts** frames selectively to the specific destination port using MAC table.
Collision Domains	All ports share a single collision domain (high collision risk).	Each individual port forms its own dedicated collision domain.
Bandwidth Sharing	Total bandwidth is shared among all connected active nodes.	Dedicated, full-rate bandwidth provided to each port (full-duplex).

Q5. What is Subnetting? Give an example with CIDR calculation. [3 Marks]

Answer:

Subnetting is the practice of logically partitioning a single large IP network into smaller, distinct subnetworks (subnets) by borrowing host bits for the network ID. This reduces broadcast domains, improves routing efficiency, and conserves IP addresses.

Example: Subnetting 192.168.1.0/24 into 4 Subnets

- To create 4 subnets: Borrow 2 bits from host portion ($2^2 = 4$ subnets).
- New prefix length: `/24 + 2 = /26` ⇒ Subnet Mask: `255.255.255.192`.
- Block Size: `256 - 192 = 64` addresses per subnet (62 usable hosts).

Subnet #	Network ID	Usable Host IP Range	Directed Broadcast IP
Subnet 1	`192.168.1.0`	`192.168.1.1 - 192.168.1.62`	`192.168.1.63`
Subnet 2	`192.168.1.64`	`192.168.1.65 - 192.168.1.126`	`192.168.1.127`
Subnet 3	`192.168.1.128`	`192.168.1.129 - 192.168.1.190`	`192.168.1.191`
Subnet 4	`192.168.1.192`	`192.168.1.193 - 192.168.1.254`	`192.168.1.255`

SECTION – B (SHORT ANSWER QUESTIONS • 7.5 MARKS EACH • ATTEMPT ANY TWO)

Q6. Describe the role of the Transport Layer in the OSI Model. [7.5 Marks]

Answer:

The **Transport Layer (Layer 4)** serves as the boundary between hardware-oriented subnet layers and software application layers, providing true **end-to-end, process-to-process reliable data delivery**.

Core Function	Detailed Mechanism & Protocol Realization
1. Service-Point (Port) Addressing	Differentiates concurrent application processes executing on the same host using 16-bit Port numbers (e.g., HTTP: 80, HTTPS: 443, DNS: 53, SSH: 22).
2. Segmentation & Reassembly	Splits massive application messages into manageable chunks called **Segments**, appends sequence numbers, and reassembles them at receiver in exact order.
3. Connection Management	Manages connection-oriented transmission via 3-way handshake (TCP) or fast connectionless datagram delivery (UDP).
4. Flow Control	Uses **Sliding Window Protocol** to prevent a fast transmitter from overwhelming a slow receiver's buffer space.
5. Error Control	Enforces end-to-end integrity using 16-bit **Checksums** and Automatic Repeat reQuest (ARQ) retransmissions for dropped/corrupted segments.

Q7. Explain Network Topologies (Star, Mesh, Bus, Ring) with diagrams and comparisons. [7.5 Marks]

Answer:

Network Topology defines the geometric and logical arrangement of communication links and nodes in a network.

1. Star Topology:

All nodes connect via dedicated point-to-point links directly to a central multiport device (Hub or Switch).

Layout: `[Node A, Node B, Node C, Node D] → [Central Switch / Hub]`

Pros: Easy to install, failure of one node does not impact others. **Cons:** Central hub is a single point of failure.

2. Mesh Topology (Full Mesh):

Every node has a dedicated point-to-point link to every other node in the network.

Link Formula: For `n` nodes, requires $\frac{n(n-1)}{2}$ physical links and `(n - 1)` I/O ports per device.

Pros: Maximum redundancy, zero traffic contention, highly fault-tolerant. **Cons:** Extremely expensive cabling and port requirements.

3. Bus Topology:

All nodes share a single common linear backbone cable (coaxial cable with terminators at both ends).

Layout: `Terminator ── [Node 1] ── [Node 2] ── [Node 3] ── Terminator`

Pros: Minimal cabling needed. **Cons:** A single break in backbone takes down entire network; hard to troubleshoot.

4. Ring Topology:

Each device connects directly to its two immediate neighbors, forming a closed continuous loop.

Layout: `Node 1 → Node 2 → Node 3 → Node 4 → Node 1 (Token Passing)`

Pros: Orderly token transmission, no collisions. **Cons:** Break in ring disrupts network (unless dual counter-rotating ring like FDDI is used).

Q8. Explain Congestion Control Algorithms (Leaky Bucket and Token Bucket). [7.5 Marks]

Answer:

Traffic shaping algorithms regulate the bursty flow of packets injected into the network to mitigate congestion:

Parameter	Leaky Bucket Algorithm	Token Bucket Algorithm
Core Operating Principle	Bursty packet arrivals enter a finite buffer; packets leak out at a **strictly constant, rigid rate**.	Tokens generate into bucket at regular rate `r` up to capacity `C`. Packets transmit by consuming tokens.
Bursty Traffic Handling	**Smooths out all bursts completely**; discards packets if bucket capacity overflows.	**Allows controlled bursts** of packets up to the accumulated token capacity `C`.
Output Transmission Rate	Strictly uniform, deterministic constant rate regardless of burstiness.	Variable rate bounded by average rate `r` and peak burst capacity.
Packet Discard Policy	Discards incoming packets when buffer memory fills up.	Discards tokens when token bucket fills; packets queue or wait for fresh tokens.

SECTION – C (LONG ANSWER QUESTIONS • 15 MARKS EACH • ATTEMPT ANY THREE)

Q9. Explain the TCP Three-Way Handshake with a sequence diagram. [15 Marks]

Answer:

The **TCP 3-Way Handshake** is the synchronization mechanism used by Transmission Control Protocol to establish a reliable, full-duplex virtual connection between client and server before data exchange begins.

TCP 3-Way Handshake Connection Sequence:

Step	Direction	Control Flags & Sequence #	Action & Purpose
Step 1 (SYN)	Client → Server	`SYN = 1` `Seq = X` (Client ISN)	Client requests connection and announces its initial sequence number X. Client enters **SYN-SENT** state.

Step 2 (SYN-ACK)	Server → Client	`SYN = 1, ACK = 1`
		`Seq = Y` (Server ISN)
		`Ack_Num = X + 1`
			Server acknowledges client's request (`Ack = X + 1`) and provides its own sequence number Y. Server enters **SYN-RCVD**.

Step 3 (ACK)	Client → Server	`ACK = 1`
		`Seq = X + 1`
		`Ack_Num = Y + 1`
			Client acknowledges server's sequence number (`Ack = Y + 1`). Connection enters **ESTABLISHED** state on both endpoints.

Significance: Guarantees both parties agree on initial sequence numbers, confirms bidirectional communication capability, and prevents old duplicate connection requests from causing false sessions.

Q10. Draw and explain the 7 Layers of the ISO/OSI Reference Model. [15 Marks]

Answer:

The **Open Systems Interconnection (OSI)** reference model is a conceptual 7-layer architectural framework developed by ISO for standardized computer communication:

Layer	Layer Name	PDU	Primary Functions & Responsibilities	Standard Protocols
Layer 7	**Application**	Data	Provides user interface and network service abstraction for software apps.	HTTP, HTTPS, FTP, SMTP, DNS, DHCP
Layer 6	**Presentation**	Data	Data syntax translation, compression, encryption and decryption (SSL/TLS).	JPEG, MPEG, ASCII, TLS/SSL
Layer 5	**Session**	Data	Dialog control, token management, session checkpointing and synchronization.	RPC, NetBIOS, PPTP
Layer 4	**Transport**	Segment	End-to-end process delivery, port addressing, flow control (sliding window), error recovery.	TCP, UDP
Layer 3	**Network**	Packet	Logical IP addressing, subnet routing, packet forwarding across networks.	IPv4, IPv6, ICMP, OSPF, BGP
Layer 2	**Data Link**	Frame	Node-to-node frame delivery, physical MAC addressing, CRC error detection, CSMA/CD.	Ethernet (802.3), Wi-Fi (802.11), PPP
Layer 1	**Physical**	Bit	Transmission of raw bitstream over physical media, signal encoding, voltages.	RS-232, RJ-45, V.35, Cables

Q11. Explain Shannon Capacity Theorem with a solved numerical problem. [15 Marks]

Answer:

Shannon's Capacity Theorem establishes the theoretical upper limit on the maximum data transmission rate (Channel Capacity C) across a noisy communication channel with white Gaussian thermal noise.

> Shannon Formula: $C = B \bullet \log_2 \&left;(1 + \frac{S}{N} \&right;)$ (in bits per second, bps)
>
> Where:
> - `C` = Maximum theoretical channel capacity in bits per second (bps)
> - `B` = Channel bandwidth in Hertz (Hz)
> - `S / N` = Signal-to-Noise Ratio (SNR as a linear power ratio)

Solved Numerical Problem:

> **Problem Statement:** A noisy telephone channel has a bandwidth of *B = 3000 Hz* and a Signal-to-Noise Ratio of SNR_{dB} *= 35 dB*. Calculate the theoretical maximum channel capacity.
>
> **Step 1: Convert SNR from decibels (dB) to linear ratio:**
>
> $$\text{SNR}_{\text{dB}} = 10 \bullet \log_{10}(S/N) = 35 \Rightarrow \log_{10}(S/N) = 3.5 \Rightarrow \frac{S}{N} = 10^{3.5} \approx 3162.28$$
>
> **Step 2: Apply Shannon's Formula:**
>
> $$C = 3000 \bullet \log_2(1 + 3162.28) = 3000 \bullet \log_2(3163.28)$$
>
> Using change of base: $\log_2(3163.28) = \frac{\log_{10}(3163.28)}{\log_{10}(2)} = \frac{3.50014}{0.30103} \approx 11.6272$
>
> $$C = 3000 \times 11.6272 \approx 34{,}881.6 \text{ bps} \approx 34.88 \text{ kbps}$$

Theoretical Channel Capacity: C ≈ 34.88 kbps

Q12. Explain CSMA/CD in Ethernet and compare with CSMA/CA in Wi-Fi. [15 Marks]

Answer:

Carrier Sense Multiple Access (CSMA) protocols require nodes to "listen before talking" on a shared broadcast transmission medium.

Comparison Feature	CSMA/CD (Collision Detection)	CSMA/CA (Collision Avoidance)
Standard & Medium	IEEE 802.3 (Wired Ethernet cables).	IEEE 802.11 (Wireless Wi-Fi radio channels).
Collision Strategy	Transmits while listening; if collision signal is detected, stops immediately.	Avoids collisions prior to transmission via reservation handshakes and delays.
Collision Response	Transmits a **Jam Signal** to notify all stations, then applies **Binary Exponential Backoff**.	Cannot detect collision while sending (half-duplex RF); infers collision if no ACK received.
Control Mechanism	Collision detection hardware listening for elevated cable voltage.	**RTS (Request to Send) / CTS (Clear to Send)** handshakes + Inter-Frame Spacing (DIFS/SIFS).

Q13. Compare Circuit Switching, Packet Switching, and Message Switching. [15 Marks]

Answer:

Switching is the methodology by which nodes communicate across multiple intermediate network nodes without direct point-to-point physical links.

Feature	Circuit Switching	Packet Switching	Message Switching
Dedicated Physical Path	**Yes**; dedicated end-to-end copper/fiber circuit reserved for session duration.	**No**; dynamic independent packet routing over shared virtual links.	**No**; whole message hops from node to node across shared lines.
Call Setup Phase	Required prior to any data transmission. Adds initial setup latency.	Zero connection setup latency (in Connectionless Datagram mode).	No setup phase needed.
Bandwidth Utilization	Low/wasteful; reserved circuit capacity is idle during conversational silence.	**Very High**; dynamic statistical multiplexing shares channel bandwidth.	Moderate; links occupied only while whole message transmits.
Store & Forward Mechanism	None; continuous real-time bitstream.	Packets buffered at each router and forwarded. Minimal latency.	Entire message buffered on intermediate disks; very high delay.
Primary Application	Traditional Landline Telephony (PSTN), cellular voice calls.	**The Modern Internet (IP)**, VoIP, streaming, web browsing.	Telegraph, Telex, historic military communication systems.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-504: Numerical Methods

Session: December Comprehensive PYQ Bank Examination | **Paper Code:** 18024 | **Complete 100% Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION – A (VERY SHORT ANSWER QUESTIONS • 3 MARKS EACH • ALL 5 COMPULSORY)

Q1. Obtain a real root of the equation $f(x) = x^3 + x^2 - 1 = 0$ using Bisection method correct to 3 decimal places. [3 Marks]

Solution:

Let $f(x) = x^3 + x^2 - 1$.

- For $x = 0$: $f(0) = 0 + 0 - 1 = -1 < 0$
- For $x = 1$: $f(1) = 1 + 1 - 1 = +1 > 0$

Since $f(0) \cdot f(1) < 0$, a real root lies in the open interval **(0, 1)**.

n	a	b	c = (a+b)/2	f(c)	Sign	Updated Interval
1	0.0000	1.0000	0.5000	-0.6250	Negative (< 0)	[0.5000, 1.0000]
2	0.5000	1.0000	0.7500	-0.0156	Negative (< 0)	[0.7500, 1.0000]
3	0.7500	1.0000	0.8750	+0.4355	Positive (> 0)	[0.7500, 0.8750]
4	0.7500	0.8750	0.8125	+0.1965	Positive (> 0)	[0.7500, 0.8125]
5	0.7500	0.8125	0.7812	+0.0872	Positive (> 0)	[0.7500, 0.7812]
6	0.7500	0.7812	0.7656	+0.0350	Positive (> 0)	[0.7500, 0.7656]
7	0.7500	0.7656	0.7578	+0.0095	Positive (> 0)	[0.7500, 0.7578]
8	0.7500	0.7578	0.7539	-0.0031	Negative (< 0)	[0.7539, 0.7578]
9	0.7539	0.7578	0.7559	+0.0032	Positive (> 0)	[0.7539, 0.7559]
10	0.7539	0.7559	0.7549	+0.00002	Positive (≥ 0)	[0.7539, 0.7549]
11	0.7539	0.7549	0.7544	-0.0015	Negative (< 0)	[0.7544, 0.7549]

At iteration 11, interval width is $|0.7549 - 0.7544| = 0.0005 < 0.001$.

Final Converged Root: x ≈ 0.755 (correct to 3 decimal places)

Q2. Define backward differences and prove the operator identity $E\nabla = \nabla E = \Delta$. [3 Marks]

Solution:

Definition: For equispaced points $x_i = x_0 + i \cdot h$, the **Backward Difference Operator (∇)** is defined as:

$$\nabla y_i = y_i - y_{i-1} \quad \text{and} \quad \nabla^k y_i = \nabla^{k-1} y_i - \nabla^{k-1} y_{i-1} \ (k \geq 2)$$

Backward Difference Table for ordered pairs (x_0, y_0) to (x_4, y_4):

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
x_0	y_0	–	–	–	–
x_1	y_1	$\nabla y_1 = y_1 - y_0$	–	–	–
x_2	y_2	$\nabla y_2 = y_2 - y_1$	$\nabla^2 y_2$	–	–
x_3	y_3	$\nabla y_3 = y_3 - y_2$	$\nabla^2 y_3$	$\nabla^3 y_3$	–
x_4	y_4	$\nabla y_4 = y_4 - y_3$	$\nabla^2 y_4$	$\nabla^3 y_4$	$\nabla^4 y_4$

Operator Proofs:

1. $E \bullet \nabla f(x) = E\,[f(x) - f(x-h)] = f(x+h) - f(x) = \Delta f(x) \Rightarrow \boldsymbol{E\nabla = \Delta}$
2. $\nabla \bullet E\, f(x) = \nabla\,[f(x+h)] = f(x+h) - f(x) = \Delta f(x) \Rightarrow \boldsymbol{\nabla E = \Delta}$

Hence Proved: $E\nabla = \nabla E = \Delta$

Q3. Find the value of $\frac{dy}{dx}$ at $x = 1.2$ for points (1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552)... [3 Marks]

Solution:

Given step size $h = 1.2 - 1.0 = 0.2$. Let $x_0 = 1.2$, $y_0 = 3.3201$.

Neighbors: $x_{-1} = 1.0$, $y_{-1} = 2.7183$ and $x_1 = 1.4$, $y_1 = 4.0552$.

Central Difference Formula (First Derivative):

$$f'(x_0) \approx \frac{y_1 - y_{-1}}{2h}$$

$$f'(1.2) = \frac{4.0552 - 2.7183}{2(0.2)} = \frac{1.3369}{0.40} = 3.34225$$

(Newton's Forward Difference check: $f'(1.2) \approx (1/0.2)[0.7351 - 0.5(0.1627) + (1/3)(0.0361)] = 3.3289$. True derivative $e^{1.2} \approx 3.3201$.)

dy/dx at x = 1.2 is 3.34225

Q4. Solve the following linear system using Gauss Elimination method: [3 Marks]

$$10x + y + z = 12$$
$$2x + 10y + z = 13$$
$$x + y + 3z = 5$$

Solution:

Step 1: Form the Augmented Matrix [A | B]:

$$\left(\begin{array}{ccc|c} 10 & 1 & 1 & 12 \\ 2 & 10 & 1 & 13 \\ 1 & 1 & 3 & 5 \end{array}\right)$$

Step 2: Forward Elimination to Upper Triangular Form:

- $R_2 \to R_2 - (2/10)R_1 = R_2 - 0.2R_1 \Rightarrow$ [0, 9.8, 0.8 | 10.6]
- $R_3 \to R_3 - (1/10)R_1 = R_3 - 0.1R_1 \Rightarrow$ [0, 0.9, 2.9 | 3.8]
- $R_3 \to R_3 - (0.9 / 9.8)R_2 \Rightarrow$ [0, 0, 2.82653 | 2.82653]

$$\left(\begin{array}{ccc|c} 10 & 1 & 1 & 12 \\ 0 & 9.8 & 0.8 & 10.6 \\ 0 & 0 & 2.82653 & 2.82653 \end{array}\right)$$

Step 3: Back Substitution:

- From Row 3: 2.82653 • z = 2.82653 ⇒ **z = 1**
- From Row 2: 9.8 • y + 0.8(1) = 10.6 ⇒ 9.8 • y = 9.8 ⇒ **y = 1**
- From Row 1: 10 • x + 1(1) + 1(1) = 12 ⇒ 10 • x = 10 ⇒ **x = 1**

Unique Solution: x = 1, y = 1, z = 1

Q5. Solve $\frac{dy}{dx} = x + y^2$, with y(0) = 1 using Picard's method up to second approximation. [3 Marks]

Solution:

Picard's Successive Approximation Formula:

$$y^{(n)}(x) = y_0 + \int_{x_0}^{x} f(t, y^{(n-1)}(t))\, dt$$

Given initial conditions: $x_0 = 0$, $y_0 = 1$, and $f(t, y) = t + y^2$.

First Approximation ($y^{(1)}$):

$$y^{(1)}(x) = 1 + \int_0^x (t + 1^2)\, dt = 1 + \left[\frac{t^2}{2} + t\right]_0^x = 1 + x + \frac{x^2}{2}$$

Second Approximation ($y^{(2)}$):

$$y^{(2)}(x) = 1 + \int_0^x \left[t + (1 + t + \frac{t^2}{2})^2\right] dt$$

Expand: $(1 + t + \frac{t^2}{2})^2 = 1 + 2t + 2t^2 + t^3 + \frac{t^4}{4}$

$$\text{Integrand} = t + 1 + 2t + 2t^2 + t^3 + \frac{t^4}{4} = 1 + 3t + 2t^2 + t^3 + \frac{t^4}{4}$$

$$y^{(2)}(x) = 1 + \left[t + \frac{3t^2}{2} + \frac{2t^3}{3} + \frac{t^4}{4} + \frac{t^5}{20}\right]_0^x$$

$$y^{(2)}(x) = 1 + x + \frac{3}{2}x^2 + \frac{2}{3}x^3 + \frac{1}{4}x^4 + \frac{1}{20}x^5$$

$y^{(1)} = 1 + x + x^2/2$ | $y^{(2)} = 1 + x + (3/2)x^2 + (2/3)x^3 + (1/4)x^4 + (1/20)x^5$

SECTION – B (SHORT ANSWER QUESTIONS • 7.5 MARKS EACH • ATTEMPT ANY TWO)

Q6. Find a real root of the equation $2x = \log_{10}(x) + 7$ between 3 and 4 using Method of False Position (Regula Falsi). [7.5 Marks]

Solution:

Let $f(x) = 2x - \log_{10}(x) - 7 = 0$.

- For $x_0 = 3$: $f(3) = 2(3) - \log_{10}(3) - 7 = 6 - 0.47712 - 7 = -1.47712 < 0$
- For $x_1 = 4$: $f(4) = 2(4) - \log_{10}(4) - 7 = 8 - 0.60206 - 7 = +0.39794 > 0$

Since $f(3)$ and $f(4)$ have opposite signs, the root lies in the interval **(3, 4)**.

Regula Falsi Iteration Formula:

$$x_2 = \frac{x_0 \cdot f(x_1) - x_1 \cdot f(x_0)}{f(x_1) - f(x_0)}$$

Iteration 1:

$$x_2 = \frac{3(0.39794) - 4(-1.47712)}{0.39794 - (-1.47712)} = \frac{1.19382 + 5.90848}{1.87506} = \frac{7.10230}{1.87506} \approx 3.78777$$

Evaluate $f(x_2)$:

$f(3.78777) = 2(3.78777) - \log_{10}(3.78777) - 7 = 7.57554 - 0.57838 - 7 = -0.00284 < 0$.

Root lies in **[3.78777, 4.00000]**.

Iteration 2:

$$x_3 = \frac{3.78777(0.39794) - 4(-0.00284)}{0.39794 - (-0.00284)} = \frac{1.50731 + 0.01136}{0.40078} = \frac{1.51867}{0.40078} \approx 3.7893$$

Evaluate $f(x_3)$: $f(3.7893) = 2(3.7893) - \log_{10}(3.7893) - 7 = 7.57860 - 0.57856 - 7 = +0.00004 \approx 0$.

Final Converged Root: x ≈ 3.7893

Q7. Find the missing terms in the following table using finite differences: [7.5 Marks]

x	0	5	10	15	20	25	30
y	1	3	? (y_2)	73	225	? (y_5)	1153

Solution:

Let the missing values be y_2 (at $x = 10$) and y_5 (at $x = 25$).

Since 5 values are given (y_0, y_1, y_3, y_4, y_6), the data can be represented by a polynomial of degree 4.

Therefore, the 5th finite difference is identically zero: $\Delta^5 y_0 = 0$ and $\Delta^5 y_1 = 0$.

Using shift operator $\Delta = E - 1$:

$$(E-1)^5 y = (E^5 - 5E^4 + 10E^3 - 10E^2 + 5E - 1)\, y = 0$$

Equation 1 (applying on y_0):

$$y_5 - 5y_4 + 10y_3 - 10y_2 + 5y_1 - y_0 = 0$$
$$y_5 - 5(225) + 10(73) - 10y_2 + 5(3) - 1 = 0$$
$$y_5 - 1125 + 730 - 10y_2 + 15 - 1 = 0$$
$$\mathbf{-10y_2 + y_5 = 381} \qquad \text{— (Equation 1)}$$

Equation 2 (applying on y_1):

$$y_6 - 5y_5 + 10y_4 - 10y_3 + 5y_2 - y_1 = 0$$
$$1153 - 5y_5 + 10(225) - 10(73) + 5y_2 - 3 = 0$$
$$1153 - 5y_5 + 2250 - 730 + 5y_2 - 3 = 0$$
$$5y_2 - 5y_5 + 2670 = 0 \Rightarrow y_2 - y_5 = -534 \Rightarrow \mathbf{y_5 - y_2 = 534} \qquad \text{— (Equation 2)}$$

Solving the Linear System:

From Equation 2: $y_5 = y_2 + 534$.

Substitute into Equation 1: $-10y_2 + (y_2 + 534) = 381 \Rightarrow -9y_2 = 381 - 534 = -153 \Rightarrow \mathbf{y_2 = 17}$.

Then $y_5 = 17 + 534 = \mathbf{551}$.

Missing Values: $y_2 = 17$ (at x = 10) | $y_5 = 551$ (at x = 25)

Q8. Evaluate $I = \int_3^7 x^2 \log_{10}(x)\, dx$ using Simpson's 1/3 rule with $n = 4$. [7.5 Marks]

Solution:

Given limits $a = 3$, $b = 7$, and number of sub-intervals $n = 4$.

Step size: $h = (b - a) / n = (7 - 3) / 4 = \mathbf{1}$.

i	x_i	Function $f(x) = x^2 \log_{10}(x)$	Ordinate y_i
0	3	$3^2 \cdot \log_{10}(3) = 9 \times 0.477121$	4.2941
1	4	$4^2 \cdot \log_{10}(4) = 16 \times 0.602060$	9.6330
2	5	$5^2 \cdot \log_{10}(5) = 25 \times 0.698970$	17.4743
3	6	$6^2 \cdot \log_{10}(6) = 36 \times 0.778151$	28.0134
4	7	$7^2 \cdot \log_{10}(7) = 49 \times 0.845098$	41.4099

Simpson's 1/3 Rule Formula:

$$I = \frac{h}{3}\left[(y_0 + y_4) + 4(y_1 + y_3) + 2(y_2)\right]$$

Sum of Extremes: $y_0 + y_4 = 4.2941 + 41.4099 = \mathbf{45.7040}$

Sum of Odd Ordinates: $4(y_1 + y_3) = 4(9.6330 + 28.0134) = 4(37.6464) = \mathbf{150.5856}$

Sum of Even Ordinates: $2(y_2) = 2(17.4743) = \mathbf{34.9486}$

Total Bracket Sum $= 45.7040 + 150.5856 + 34.9486 = \mathbf{231.2382}$

$$I = \frac{1}{3} \times 231.2382 = \mathbf{77.0794}$$

Evaluated Integral: $I \approx 77.0794$

SECTION – C (LONG ANSWER QUESTIONS • 15 MARKS EACH • ATTEMPT ANY THREE)

Q9. Use Newton-Raphson method to find a real root of $\sin(x) = x / 2$ correct to 3 decimal places between π/2 and π. [15 Marks]

Solution:

Let $f(x) = \sin(x) - \frac{x}{2} = 0$.

Differentiating with respect to x: $f'(x) = \cos(x) - 0.5$.

Given interval: [π/2, π] ≈ [1.5708, 3.1416] radians.

Let initial approximation be $\mathbf{x_0 = 2.0}$ **radians**.

Newton-Raphson Iteration Formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Iteration 1 (at $x_0 = 2.0$):

$$f(2.0) = \sin(2.0) - 1.0 = 0.90930 - 1.0 = \mathbf{-0.09070}$$
$$f'(2.0) = \cos(2.0) - 0.5 = -0.41615 - 0.5 = \mathbf{-0.91615}$$
$$x_1 = 2.0 - \frac{-0.09070}{-0.91615} = 2.0 - 0.09900 = \mathbf{1.90100}$$

Iteration 2 (at $x_1 = 1.90100$):

$$f(1.90100) = \sin(1.90100) - \frac{1.90100}{2} = 0.94598 - 0.95050 = \mathbf{-0.00452}$$
$$f'(1.90100) = \cos(1.90100) - 0.5 = -0.32422 - 0.5 = \mathbf{-0.82422}$$
$$x_2 = 1.90100 - \frac{-0.00452}{-0.82422} = 1.90100 - 0.00548 = \mathbf{1.89552}$$

Iteration 3 (at $x_2 = 1.89552$):

$$f(1.89552) = \sin(1.89552) - \frac{1.89552}{2} = 0.94775 - 0.94776 = \mathbf{-0.00001}$$
$$f'(1.89552) = \cos(1.89552) - 0.5 = -0.31899 - 0.5 = \mathbf{-0.81899}$$
$$x_3 = 1.89552 - \frac{-0.00001}{-0.81899} = 1.89552 - 0.00001 = \mathbf{1.89549}$$

Root: x ≈ 1.895 radians (or ≈ 108.57°)

Q10. Find the Lagrange interpolating polynomial of degree 2 for *y = $\log_e(x)$* from the given table. Hence determine *$\log_e(2.7)$*. [15 Marks]

x	2.0	2.5	3.0
y = $\log_e$(x)	0.69315	0.91629	1.09861

Solution:

Given $x_0 = 2.0, y_0 = 0.69315$; $x_1 = 2.5, y_1 = 0.91629$; $x_2 = 3.0, y_2 = 1.09861$.

Target value: $x = 2.7$.

Lagrange Interpolation Formula for n = 2:

$$P_2(x) = L_0(x)\, y_0 + L_1(x)\, y_1 + L_2(x)\, y_2$$

Calculating Lagrange Cardinal Polynomials at x = 2.7:

$$L_0(2.7) = \frac{(2.7-2.5)(2.7-3.0)}{(2.0-2.5)(2.0-3.0)} = \frac{(0.2)(-0.3)}{(-0.5)(-1.0)} = \frac{-0.06}{0.50} = \mathbf{-0.12}$$
$$L_1(2.7) = \frac{(2.7-2.0)(2.7-3.0)}{(2.5-2.0)(2.5-3.0)} = \frac{(0.7)(-0.3)}{(0.5)(-0.5)} = \frac{-0.21}{-0.25} = \mathbf{+0.84}$$
$$L_2(2.7) = \frac{(2.7-2.0)(2.7-2.5)}{(3.0-2.0)(3.0-2.5)} = \frac{(0.7)(0.2)}{(1.0)(0.5)} = \frac{0.14}{0.50} = \mathbf{+0.28}$$

Verification: $L_0 + L_1 + L_2 = -0.12 + 0.84 + 0.28 = 1.00$ ✓

Evaluating $P_2(2.7)$:

$$P_2(2.7) = (-0.12)(0.69315) + (0.84)(0.91629) + (0.28)(1.09861)$$
$$= -0.083178 + 0.769684 + 0.307611$$
$$= \mathbf{0.994117 \approx 0.99412}$$

(Exact analytical value: ln(2.7) ≈ 0.99325; error ≈ 0.00087.)

Interpolated Value: $\log_e(2.7) \approx 0.99412$

Q11. Use Simpson's 3/8 rule to evaluate $\int_0^1 \frac{dx}{1+x}$ with $h = 1/6$. [15 Marks]

Solution:

Given $a = 0$, $b = 1$, step size $h = 1/6 \Rightarrow n = (1 - 0)/(1/6) = 6$ subintervals.

Integrand: $f(x) = \frac{1}{1+x}$.

i	x_i	Exact $y_i = 1 / (1 + x_i)$	Decimal Value
0	0	1 / 1	1.000000
1	1/6	1 / (7/6) = 6/7	0.857143
2	2/6	1 / (8/6) = 6/8 = 3/4	0.750000
3	3/6	1 / (9/6) = 6/9 = 2/3	0.666667
4	4/6	1 / (10/6) = 6/10 = 3/5	0.600000
5	5/6	1 / (11/6) = 6/11	0.545455
6	1	1 / 2	0.500000

Simpson's 3/8 Rule Formula:

$$I = \frac{3h}{8}\left[(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3)\right]$$

Extremes: $y_0 + y_6 = 1.000000 + 0.500000 = \mathbf{1.500000}$

Non-multiples of 3: $3(y_1 + y_2 + y_4 + y_5) = 3(0.857143 + 0.750000 + 0.600000 + 0.545455) = 3(2.752598)$
$= \mathbf{8.257794}$

Multiples of 3: $2(y_3) = 2(0.666667) = \mathbf{1.333334}$

Sum of brackets $= 1.500000 + 8.257794 + 1.333334 = \mathbf{11.091128}$

$$I = \frac{3(1/6)}{8} \times 11.091128 = \frac{1/2}{8} \times 11.091128 = \frac{11.091128}{16} = \mathbf{0.693196}$$

(Exact analytical integral: $\int_0^1 dx/(1+x) = [\ln(1+x)]_0^1 = \ln(2) \approx 0.693147$. Highly accurate to 4 decimal places.)

Evaluated Integral: I ≈ 0.693196

Q12. Solve the following linear system using Gauss-Seidel iterative method: [15 Marks]

$$6x + y + z = 20$$
$$x + 4y - z = 6$$
$$x - y + 5z = 7$$

Solution:

Step 1: Check Diagonal Dominance Condition:

- Equation 1: $|6| > |1| + |1| = 2$ (Dominant in x ✓)
- Equation 2: $|4| > |1| + |-1| = 2$ (Dominant in y ✓)
- Equation 3: $|5| > |1| + |-1| = 2$ (Dominant in z ✓)

The system is strictly diagonally dominant. Convergence is guaranteed.

Step 2: Express in Iterative Form:

$$x^{(k+1)} = \frac{20 - y^{(k)} - z^{(k)}}{6}$$
$$y^{(k+1)} = \frac{6 - x^{(k+1)} + z^{(k)}}{4}$$
$$z^{(k+1)} = \frac{7 - x^{(k+1)} + y^{(k+1)}}{5}$$

Step 3: Iteration Table (Initial Approximation: $x^{(0)} = 0$, $y^{(0)} = 0$, $z^{(0)} = 0$):

Iteration (k)	x = (20 – y – z) / 6	y = (6 – x + z) / 4	z = (7 – x + y) / 5
1	(20 - 0 - 0)/6 = **3.3333**	(6 - 3.3333 + 0)/4 = **0.6667**	(7 - 3.3333 + 0.6667)/5 = **0.8667**
2	(20 - 0.6667 - 0.8667)/6 = **3.0778**	(6 - 3.0778 + 0.8667)/4 = **0.9472**	(7 - 3.0778 + 0.9472)/5 = **0.9739**
3	(20 - 0.9472 - 0.9739)/6 = **3.0131**	(6 - 3.0131 + 0.9739)/4 = **0.9902**	(7 - 3.0131 + 0.9902)/5 = **0.9954**
4	(20 - 0.9902 - 0.9954)/6 = **3.0024**	(6 - 3.0024 + 0.9954)/4 = **0.9983**	(7 - 3.0024 + 0.9983)/5 = **0.9992**
5	(20 - 0.9983 - 0.9992)/6 = **3.0004**	(6 - 3.0004 + 0.9992)/4 = **0.9997**	(7 - 3.0004 + 0.9997)/5 = **0.9999**
6	**3.0000**	**1.0000**	**1.0000**

Converged Solution: x = 3, y = 1, z = 1

Q13. Using Fourth-Order Runge-Kutta method, find *y(0.1)* and *y(0.2)* correct to 4 decimal places, given: [15 Marks]

$$\frac{dy}{dx} = y - x, \quad \text{with initial condition } y(0) = 2$$

Solution:

Given $f(x, y) = y - x$, initial point $x_0 = 0$, $y_0 = 2$, step size $h = 0.1$.

Part 1: Calculation of y(0.1) from $x_0 = 0$, $y_0 = 2$:

$$k_1 = h \bullet f(x_0, y_0) = 0.1(2 - 0) = \mathbf{0.200000}$$
$$k_2 = h \bullet f(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}) = 0.1 \bullet f(0.05, 2.10) = 0.1(2.10 - 0.05) = \mathbf{0.205000}$$
$$k_3 = h \bullet f(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}) = 0.1 \bullet f(0.05, 2.1025) = 0.1(2.1025 - 0.05) = \mathbf{0.205250}$$
$$k_4 = h \bullet f(x_0 + h, y_0 + k_3) = 0.1 \bullet f(0.1, 2.20525) = 0.1(2.20525 - 0.1) = \mathbf{0.210525}$$
$$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$
$$= \frac{1}{6}(0.200000 + 0.410000 + 0.410500 + 0.210525) = \frac{1.231025}{6} \approx \mathbf{0.205171}$$
$$y(0.1) = y_0 + k = 2 + 0.205171 = 2.2052$$

Part 2: Calculation of y(0.2) from $x_1 = 0.1$, $y_1 = 2.205171$:

$k_1 = h \bullet f(x_1, y_1) = 0.1(2.205171 - 0.1) = \mathbf{0.210517}$

$k_2 = h \bullet f(0.1 + 0.05, 2.205171 + 0.105258) = 0.1 \bullet f(0.15, 2.310430) = 0.1(2.310430 - 0.15) = \mathbf{0.216043}$

$k_3 = h \bullet f(0.1 + 0.05, 2.205171 + 0.108022) = 0.1 \bullet f(0.15, 2.313193) = 0.1(2.313193 - 0.15) = \mathbf{0.216319}$

$k_4 = h \bullet f(0.1 + 0.1, 2.205171 + 0.216319) = 0.1 \bullet f(0.2, 2.421490) = 0.1(2.421490 - 0.2) = \mathbf{0.222149}$

$$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$= \frac{1}{6}(0.210517 + 0.432086 + 0.432638 + 0.222149) = \frac{1.297390}{6} \approx \mathbf{0.216232}$$

$$y(0.2) = y_1 + k = 2.205171 + 0.216232 = 2.4214$$

y(0.1) = 2.2052 | y(0.2) = 2.4214